

★ CUBE ★

Quant | Ch06 3D Mensuration + DI | Topic 1 of 12

► Meaning in one line

Side³ , 6×side² , space-diagonal.

► Key points

Point	Example
Volume/surface/diagonal: $a^3 + 6a^2 + a\sqrt{3}$.	$a5 \rightarrow 125, 150, 5\sqrt{3}$.
Use: paint + cut.	painted + cut $n = \text{faces} \times n^2 \dots$ (DI link next).

★ IMPORTANT FOR EXAM

Cuboid + cylinder + cone

⚠ TRAP - read twice

Area-for-volume + 'diagonal = side' + units-mix

✓ **Try yourself:** Cube 6 volume + surface? **Answer:** 216 + 216.

★ CUBOID ★

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► Meaning in one line

$L \times W \times H$, $2(LW, WH, HL)$, space-diagonal.

► Key points

Point	Example
Volume/surface/diagonal.	$6 \times 4 \times 3$: V72, S108, $d\sqrt{61}$.
Use: rooms + tanks + boxes.	paint-4-walls = $2h(L+W)$.

★ IMPORTANT FOR EXAM

Cylinder + cone + sphere

⚠ TRAP - read twice

Cube-formula + 'all sides' + diagonal-miss + units

✓ **Try yourself:** $4 \times 3 \times 5$ volume + surface? **Answer:** 60 + 94.

★ CYLINDER ★

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► Meaning in one line

$\pi r^2 h$, $2\pi r(h, r)$, curved.

► Key points

Point	Example
Volume/CSA/TSA.	$r7\ h10$: V1540, CSA440, TSA748.
Use: pipes + tanks + hollow.	hollow = outer–inner.

★ IMPORTANT FOR EXAM

Cone + sphere + hemisphere

⚠ TRAP - read twice

Circle-area + 'height = slant' + diameter-radius

✓ **Try yourself:** $r14\ h20$ (22/7) volume? **Answer:** $22/7 \times 196 \times 20 = 12320$.

★ CONE ★

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► Meaning in one line

Third- π - r^2h , π - r - l , slant.

► Key points

Point	Example
Volume/CSA + slant.	r6 h8: l10, V301.6, CSA188.5.
Use: funnels + tents + hats + frustum.	-

★ IMPORTANT FOR EXAM

Sphere + hemisphere + conversions

⚠ TRAP - read twice

Cylinder-volume + 'height = slant' + third-miss

✓ **Try yourself:** r9 h12 (3.14) volume? **Answer:** $1017.36 \approx 1017$ (third $\times$ 81 $\times$ 12 $\times$ 3.14).

★ SPHERE ★

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► Meaning in one line

Four-thirds- π - r^3 , 4π - r^2 , diameter.

► Key points

Point	Example
Volume/surface.	r21: V38808, S5544.
Use: balls + hemispheres.	hollow = outer–inner.

★ IMPORTANT FOR EXAM

Hemisphere + conversions + melting

⚠ TRAP - read twice

Circle-area + '4/3-miss' + diameter-radius

✓ **Try yourself:** r10.5 (22/7) volume? **Answer:** $\frac{4}{3} \times \frac{22}{7} \times 1157.6 \approx 4851$.

★ HEMISPHERE + VOLUME-CAPACITY ★

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► Meaning in one line

Two-thirds- π - r^3 , 3π - r^2 , bowls , domes.

► Key points

Point	Example
Hemi volume/surface.	r7: V718.7, S462.
1 m ³ = 1000 L.	tank 2×1.5×1 m = 3000 L.

★ IMPORTANT FOR EXAM

Melting + DI-tables

⚠ TRAP - read twice

Full-sphere + 'litre = m³' + factor-1000-miss

✓ **Try yourself:** Hemisphere r14 vol (22/7)? Tank 3×2×1.5 litres? **Answer:** 5745.3; 9000 L.

★ MELTING/RECASTING (VOLUME CONSERVED) ★

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► Meaning in one line

Melt , recast , equate , solve.

► Key points

Point	Example
Equate volumes + solve.	cube $6^3 = 216 \rightarrow$ small $2^3 = 8$ each $\rightarrow 27$ cubes.
Wires/sheets.	melt sphere $r_6 \rightarrow$ wire r_1 : length $= 288\pi/\pi = 288$.

★ IMPORTANT FOR EXAM

DI tables + bars + pies

⚠ TRAP - read twice

Surface-equate + 'melt = average' + units-mix

✓ **Try yourself:** Cube 9^3 into 3^3 cubes count? **Answer:** $729/27 = 27$.

★ TABLE DI ★

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► Meaning in one line

Read , total , average , ratio , percent , max/min , difference.

► Key points

Point	Example
Read + compute.	total + share + growth.
Max/min/difference + footnotes + units.	-

★ IMPORTANT FOR EXAM

Bar + pie + line + mixed

⚠ TRAP - read twice

Eyeball + 'approx fine' + units/footnotes-missed

✓ **Try yourself:** Table + share + growth + total. Steps? **Answer:** Read + total + ratio + percent + max.

★ BAR GRAPH DI ★

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► Meaning in one line

Height , value , compare , total , average , percent , trend.

► Key points

Point	Example
Read + value + compare + total.	-
Scale + grouped/stacked + legend.	-

★ IMPORTANT FOR EXAM

Pie + line + mixed + approx

⚠ TRAP - read twice

Eyeball + 'taller = double' + scale-missed + legend-ignored

✓ **Try yourself:** Bar + scale + legend + compare + total. Steps? **Answer:** Scale + value + compare + percent.

★ PIE CHART DI ★

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► Meaning in one line

360°, percent, value, sector, compare.

► Key points

Point	Example
Degrees/share/total.	25% of 360 = 90°; total 7200 → 1800.
Compare + legend.	A:B sectors 120:90 = 4:3.

★ IMPORTANT FOR EXAM

Line + mixed + approx

⚠ TRAP - read twice

Eyeball-angles + 'biggest = half' + total-missed + no-compute

✓ **Try yourself:** 30% sector + total 5400 value? Ratio 72°:108°? **Answer:** 1620; 2:3.

★ LINE GRAPH DI ★

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► Meaning in one line

Slope , trend , peak/trough , compare , percent , forecast.

► Key points

Point	Example
Read + slope + peak + compare.	-
Axes + legend + units + forecast.	-

★ IMPORTANT FOR EXAM

Mixed + approx + eliminate

⚠ TRAP - read twice

Eyeball-trend + 'rising = good' + axes-ignored + no-compute

✓ **Try yourself:** Line + peak + growth% + forecast. Steps? **Answer:** Axes + values + percent + trend.

★ MIXED DI + APPROXIMATION ★

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► Meaning in one line

Table, bar , pie, line , cross.

► Key points

Point	Example
Mixed + cross-verify.	table total vs pie share.
Round + eliminate.	options far → estimate + strike.

★ IMPORTANT FOR EXAM

Full Quant shelf: 73pp + compendium + mocks + revision!

⚠ TRAP - read twice

One-source + 'options exact' + no-cross/round + no-eliminate = slow + wrong + fail! Cross + round.

✓ **Try yourself:** Options 1.2L/1.5L/1.8L, estimate 1.52L. Pick + why? **Answer:** 1.5L — nearest + eliminate.